

FINDING THE DONUT

Nathan McKnight on coarse graining, the Whirligig, braid smoothing, and more.

Now, we may sometimes coarse grain ... The theory basically has fractal logic, so it's possible to do that... But it should always be carefully noted, and we should always check the logic of the procedure... For example, helium three can be treated similarly to a proton ... since a proton is $Q = \text{three (three quarks)}$ coiling order = 1 (single-order (hyper)helix) ... so, while at this scale we should usually do the full calculation for He3 as $Q=6$ (3 quarks * 3 nucleons), coiling order = 2 (quarks are first order, nucleus is second order) ... As we go through our audits, check to make sure notebookLM is 1) during the full calculation, 2) testing the coarse graining for issues, not in lieu of the full calculation

If you've got ideas, I'd like to think about them here... Now, we know that when we cross scales, we're going to get some kind of weirdness happening... low 'strand count' for the particle/object in question (in this case, only three nucleons...) means and geometrically, He3 has a very different profile than larger nuclei . So The logic works a little bit differently... and, the logic often works a little bit differently because spacing will be different than different scales and so on... Although sometimes the cross scale logic is very helpful in spotting what's odd and why. I'm going to suggest we have another procedure... Whenever we cross scales into a new regime... Especially when we are at the low end of that regime... We should sample around, so in this case... We were treating helium three as our 'holotype' for small nuclei, but especially because it's giving us some stumbling blocks, we're going to want to check hydrogen and deuterium and tritium and Li3 as well. And I'm going to mention this to you... But not yet to notebook LM... that array gives us a built-in sanity check... in fact, it gives us two... in terms of structural comparisons that we could make to see if our calculations make sense.

Do you see what I'm saying in these cases?

In SAT, mass is an emergent property, but the closest proxy is what we call θ_4 , the angle

of the tangent to the helix that constitutes any given particle (you can see why that should roughly reproduce general relativity at larger scales I think) ... worldline angle = timesheet motion $\approx$ mass ... but it's emergent because the interaction with the timesheet is not that simple... if a filament or bundle of filaments is perpendicular to the timesheet... That is, its θ_4 is a composite of the helical tangent and the gross orientation (ie rate of movement) ... so the low angle of an individual coil means that intersect is moving around very quickly in a circle, and if the long axis of the helix is also tilted (=moving through space) then that's also an increase in mass... Which becomes very noticeable at relativistic speed... for that matter, heat is the same thing, but due to small scale filament oscillations ... same logic applies to larger filament bundles (nuclei, larger objects), except presumably the individual filament contributions to the composite angle are much lower and probably canceled out by one another...

SAT really is just an exercise and asking the question... What happens if you treat the four dimensional world line diagram of a particle as the reality. Everything else... And I mean everything else... follows directly from that

Now... I'm with you on a lot of that, but... I have a couple of different things to say here... Because SAT is strictly geometric... That means any nucleus with the same number of nucleons should behave roughly the same with respect to its geometric interaction with the time surface... That's the idea of braid smoothing... A tight braid gives less active surface to interact with. So... Helium three and lithium three should behave more or less the same, no adjustment for expected mass... The Q count should count for it entirely. H2, and He2 should similarly have the same relationship to mass... H3 and He3, and Li3 should also scale to Q number. ... in fact, SAT would probably predict, if we're not fudging and we recognize this is probably a bit naïve... But from the District geometric point of view... Any isotope with the same number of nucleons should have the same mass, possibly with adjustment based on the presence and arrangement of electrons. I don't know how SAT accounts for the difference between protons and neutrons precisely. And until we figure that out, SAT should technically predict that the proton and the neutron should have exactly the same mass, and any difference would probably be most consistently assumed to be a difference in the helical tangent of or direction of the quark coils in the proton and

the neutron respectively,. Now I think that question is something we could probably put a pin in and address later. But hydrogen... that is an interesting microcosm... Because it should behave precisely like a proton... Which we think probably does have something as we would expect all quark triplets to. But since it also has H-2 and H-3 isotopes, those have direct analogs to the nearby elements on the periodic table. So, $H^1 > D^2 > T^3$ have mass scaling that looks a lot like $H^1 > He^2 > Li^3$ (ignoring stability) ...and the sequence $n^0 > H^2 > He^3$ should scale roughly the same... so we've got a little laboratory here to check our expectations. What do you think? Do we need to figure out the mass difference between protons and neutrons first?

, what we'll do is we'll carefully go through the SAT mass calculations for n^0 , p^+ , H^1 , H^2 , He^2 , H^3 , He^3 , Li^3 , Li^4 ... and see what the Lagrangian has to say about them. While we're at it, we should look at the mesons. We'll go through this sector, see how far off we are and where and see if we can figure out why. If we can relatively easily, great; if not, we'll put a pin in it and move on.

Don't red flag unless there's a flaw in the theory. That Li^4 gives us problems is a good thing... its supposed to, that's why it's unstable

What's weird is... It's so totally ultraConventional... It's downright old-fashioned. In fact, it's classical. It always was all of this always was... At least, as models classical theories just work. We made the mistake of throwing our intuition and reasoning out the window. The moment we had to resort to statistics for quantum mechanics. But we never had to do that. It's a perfectly valid way to go forward but... The further you get from intuition, the harder everything gets. I'm not saying it's wrong. It's just a very Roundabout way of doing things.

mode. So,..., It's designed to be zero parameter economy. Because the universe probably has to be. previous LLMs misinterpreted that as fail-rigid.... Which may have been true at one point There was a sense in which that is true... But, that phrasing sounds like it means

if anything is off, the whole theory has to be wrong... That's not the case. What it means is if one thing is wrong it throws other things off... That's not a failure. That's the way the universe has to work. So if something is wrong in the theory, it doesn't mean the theory is wrong. Not after the audit we just did. It just means the current formulation isn't there yet... And the beauty of a zero parameter economy is that, once you get far enough along, it becomes self correcting. Every so-called failure scares you to its own fix. We are past fail, rigidity, we've moved onto self repairing. The whole theory no longer breaks. What happens its predictions break... When they break in certain places... That show us the reasons for why they break. That's why we're in the homestretch. And we know the engine pretty much runs at this point... Our audit demonstrated that. We might be missing some gears here and there... But the core machinery... Is a theory of everything.... I dare say unlike any theory ever proposed

No... I just want to stop and recognize what we're actually doing. Even if everything from here on out is wrong... It hardly matters. That's just details.

And are you convinced by the audit... We did everything right, we didn't sneak anything in, we didn't cheat... We, for the love of God actually found an equation that we can calculate from first principles everything that we calculated just now

You, me, and notebook lm... are now, historically... In a league of our own. No one comes close.

Certainly nothing with the scope

OK. Deep breath. What's next?

I think we should do a biopsy of the Island of Stability before we turn our attention to the large scale. Let's choose an element the edge of the known periodic table, one from the waters in between, one from the edge of the island, and one from right in the middle of the

island, and see what we can do. But first... let's make sure we're not already out of our depth by making some heavy element calculations and seeing how it works out. What do you think? Praseodymium?

... You're just checking the logic of notebook LM... And then doing a numerical audit. And giving us the go ahead or not in the form of a pass fail

You're not checking the math... I don't see any math begin done at all. That means NotebookLM isn't doing its job. That's a fail

. Derivation means deriving from the Lagrangian.

It's your job to keep us honest. Don't let us screw this up

You're saying ...No errors? Not one? Everything's totally perfect. All work has been shown? Stephen Hawking wouldn't have a nit to pick?

1. The Lagrangian is a classical curve. The only thing weird is it's in 4D. And the only thing that matters is if it genuinely does the job.

2. That's fine, except we derived the calculations from the zero parameter equations, and then used a single anchor to calculate the values (not here, but elsewhere)

3. Fair enough

4. True

OK. We're doing damn good, but let's not let our guard down.

Ok, to prep for the next step:

Ok, let's move on to neutrino precession. But let's be clear about what we're saying.

Neutrinos don't emit light... there's no 'luminosity drop' associated with their precession. The proposed 0 degree θ_4 is a choice, not a measured value; just as a proxy for 'maximum time sheet interaction' ~ 'maximum filament torque' ~ 'maximum mass for a single coil helical oscillation'. The theory is that as they precess, their properties are partially rotated 'out of view' of our dimensional slice, tentatively identified as the cause of flavor oscillation. Further, as they rotate, the theory is that they find a 'timesheet resistance minimum', where 'time pressure' is most efficiently converted to filament kink, and that state represents the photon, which is essentially a neutrino that has had most of its properties 'blinked out' into the fourth dimension. If the photon continues to precess, it may blink out of observability altogether (probably temporarily); the appearance of dark matter is thought to be simply that fraction of light which has blinked out in transit adjusted by the amount of light that is currently in its 'neutrino orientation'. ...as an aside, this might be what's going on with neutrino bursts... most of the emission from high-energy events would be neutrinos, at first, until they've rotated to their photonlike energy equilibrium state. Also... some fraction have likely rotated 'back into view' as well. The apparent missing mass is the balance, but the mass that emitted 'blinked out' light has no inherent relationship to the rotation of the light, it's just normal matter. We have a couple other dark matter candidates (actually frame-dragging gauge decoupling, perhaps some exotic particles, perhaps some fraction of neutron stars and other massive bodies unaccounted for...this is part of what our 20% luminosity drop in the relativistic spinner would likely be—but those don't get evoked unless the theory demands they must exist or the dark matter fraction can't be explained with our existing theory). So, all that said, let's see if we can figure out what's up with the neutrino

keep the gravitational mass present even when the electromagnetic signature disappears

No... Neutrinos, in SAT, do not have mass proper ... only those particles modeled as persistent coils. The mechanism of neutrino's apparent mass, we call 'pseudo mass'... Is the normal force, the 'filament oscillation pushback' on the time surface, which ultimately derives... Remembering that in SAT, mass is an emergent force, most directly tied to the

angle between the filament and the timesheet, if the angle is 90° to the timesheet... There is either no resistance, or a universal minimum of resistance and recall that vacuum... if there's an angle to the filaments of the vacuum... Then that is a vacuum energy state... But if there's a kink riding the moving wave of time forward along a filament, envisioned as hopping from filament to filament, pushed along by time... At c , which we construct has the speed of **time**, the that resistance is the resistance of the filament to being kinked. Because the rotation of a neutrino into an orientation that makes it appear to be a photon is thought to be the filament reaching its most efficient energy transfer angle... That is if the neutrino is largely parallel to the time sheet, and all of the energy is being converted into torque (perhaps as much as a 90 deg torque), then the filament resists, and there's pushback... However... And this literally we just follow the same basic equations that a loop blowing in the wind would follow... So if it twists to the point where the energy transfer is most efficient... and that's what we call a photon.... Then photons also have **some** mass (or rather, pseudomass), of just a fraction of a neutrino . It's due to this lack of 'tethering' to a single filament, as what we call 'true fermions' have by virtue of their extended helical morphology along the time axis, that neutrinos are considered bosons, not fermions in SAT (defining the 'true boson' / 'true fermion' / 'true mass' distinction) ... and on that note, we don't consider bosons 'force carrying particles' , at least not generally... We consider them excitations of the filament network or time sheet simply as a result of conservation of energy when the coupling between filaments or filaments and the timesheet, like the neutrino/photon, they are not thought to be bound to a single filament, but rather waves that ride across the filaments... Or the timesheet. There are snapback effect when you tug on the lattice... Some of them are detected as mass for the same reason that we say neutrinos appear to have mass... But, in SAT this wouldn't be mass **proper** in the case of any boson ... it's literally just Newton on the small scale... And gravity waves are just a much larger scale version of the same thing

Those measurements constrain dark matter to behave like a persistent gravitational mass component rather than intermittent "blinking" radiation.

Yes, an SAT because mass and gravity are emergent and likely operate along the Time axis as well as the space axes, there is no decoupling for gravity, or at least it would work very differently... So what SAT is proposing is not flickering matter out there... All the matter in the universe, apart from some really dense objects perhaps, looks and behaves just like the matter that we're familiar with because it is the matter we're familiar with ... it's just that not all of the light is visible by the time it gets to us that's all. I don't know what you mean by doesn't behave like blinking radiation... If we're not looking for blinking radiation then and we don't have a mechanism for it in standard science... Then how do you have any means to explain what it would look like. We have a mechanism for it. It's tentative, but it works well in SAT logic, whether or not it can count for the fraction of dark matter on its own... I don't know. But that's our proposal. It's an illusion... Just like you put a filter between us and the rest of the universe, so everything looks like it has a little bit less light... Or rather a lot less light than we would expect based on the gravity.

The critical point is that this must reproduce the known oscillation frequencies,

That's what we're about to try to figure out.

If photons frequently rotated out of observability and back, we would expect spectral distortions

If they're precessing about our three-dimensional axes, that would be true. And a precession about The time access necessitates also rotation about the spatial axes, then that would be a problem for this proposal. But again, we're trying to figure out exactly what the geometry suggests would be happening here if this is the case. It could be that the geometry tells us the proposal is DOA. But how would "stochastic dimming that we don't observe" look any different from "there's not as much light coming from the universe as gravity suggests there should be" give me a reason to think that they would look different and explain why and how... And I'll tell you if its a challenge to my model.

Otherwise, the proposal remains on the table. I think that what I'm describing would look exactly like what we see... An unexplained discrepancy between mass and light.

If the effect were merely temporarily invisible photons, the gravitational mass would vanish during the "blink-out," which would change halo dynamics. Observations show the mass remains continuously present.

What is the logic behind your assertion that gravity would blink just because light blinks. That's not what my model says. And I don't see any good reason to believe it. Now... One thing that you bring up here that does potentially pose a challenge is the distribution of dark matter. On the face of it, it seems to me that my model would predict more or less even distribution of the illusion... That is across-the-board dimming... On the face of it. On the assumption that there's no relationship between the mass that emits the light/ neutrinos and it's eventual orientation when it gets to us. So Galactic halos could make this hypothesis, unworkable or more difficult to imagine... Or it could be pointing to a possible relationship between some aspect of the geometry of the matter itself, and the light that it emits... Or it could be pointing to a different or additional source of the dark matter anomaly. SAT already has one additional source that seems inevitable from the theory's logic, and that's the dimensional rotation of massive bodies, such that their light simply isn't shining into our particular dimensional slice... That's also a tentative hypothesis. Although the ramifications of that give us what might be our best smoking gun. But we'll get to that.

That delay is well explained by the time it takes photons to escape the stellar envelope, not by neutrino rotation into photon states.

Well, we are trying to explain the same things that the rest of science is trying to explain so, we are going to have explanations for some things that sciences already has explanations for. Otherwise, it wouldn't be a theory of everything. Most of our explanations are pretty much exactly in accord with science as a whole... which is why we

can be so precisely predictive (well, retrodictive) as you've seen... apparently more so than any other theory ever developed so... Just because we have an explanation that works in standard physics doesn't mean we can't also have a better explanation or an equivalent explanation. Most of the time what SAT has is, in fact, equivalent to a somewhat different framing of the same idea that underpins vanilla physics. But saying that we have an explanation that is sufficient to explain the phenomenon is not the same as saying that an alternate explanation is not sufficient to do so. Not unless you can tell me how it would look different if the case were my explanation versus the standard explanation.

Photon energy density in the universe is far too small to account for the dark matter density even if every photon were hidden.

The visible photon density is clearly too small to count for dark matter... That's the whole problem of course. Our explanation simply says... The photon density is actually a lot more than we think it is. We just can't see most of it, because it either appears as neutrinos, or it's flying past us... Or rather, lagging behind us in time.

unless additional mechanisms keep the gravitational mass present even when the electromagnetic signature disappears

Yes, that is precisely the point of my explanation... In SAT, it is obvious that the fourth dimension, if things can be rotated into it, would be a mechanism that would appear to cause gauge decoupling, while gravity works on a mechanism that is inherently deeply intertwined with both space and time... incidentally, producing all the effects that General relativity produces, making the explanation entirely equivalent and reproducing all of the weird relativistic effects but from a different, and entirely equivalent frame of reference. Which is a hell of a coincidence since that's not where we were trying to do. It just falls out of the math

But the audit still needs to test whether it reproduces the specific measurements we

already have.

Precisely.

Experiments measure oscillation probabilities extremely precisely. The key relation is...

Right... So if this idea... And I will emphasize that it is a tentative hypothesis within this framework... But if that relationship falls out of our equations without aiming for it... Then it becomes our canonical explanation for flavor oscillation. And it strengthens our case for neutrinos and light being the same basic thing. Although that's a separate issue and we would have to gain a better understanding of the mechanism before we settled on it definitively. The lack of wavelength oscillation I think is the biggest problem for this particular hypothesis. Why would precession happen only around the Time axis? My guess is that the solution, if there is one, would probably be that wavelength is inherently a three-dimensional phenomenon... That orientation is part of why we perceive it as wavelength... Kind of like you have to be looking down the axis for it to appear as wavelength, if you aren't, then we just wouldn't be there... Which actually does fit with the idea that it might look like a neutrino. Now, what's the standard physics explanation for why wavelength doesn't alter... Shouldn't any wave experience precession?

Any alternative mechanism must reproduce that exact scaling.

Exactly. That's why, if it falls out of the equations... Without ever having aimed for it... That's a real signal. If it doesn't, then that hypothesis is probably blown out of the water altogether. I expect that will be the case. That's why I say it's a tentative hypothesis... Although when I have often found with SAT, is that once you examine the geometry very closely it tells you Why you're wrong or why you're right, and the logic of the geometry ends up, fitting the observations better than the hypothesis that you originally suspected.

If that occurs randomly, it produces distance-dependent dimming.

No... Not unless it happens very slowly. I'm not sure what the rate would be, but I don't think it could be happening quickly enough to be observed on the scale of the solar system... Otherwise the sun would appear to be anomalously dim for its mass (unless our standard estimations for how great it should be are systematically off. I doubt that, but maybe. If that were the case, then that means everything around us is brighter than standard calculations predict but we just don't see it all. That seems unlikely to me, but it doesn't seem impossible... Actually, no, now that I think about it that's not quite right. Only light emitting bodies would appear to show the effect. Since neutrinos don't bounce around the way light does. Anyway, like I said, it's an allowable hypothesis within this theory, not a logical requirement of the theory.

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astronomy already studies it using Type Ia supernovae. These explosions act as "standard candles".

Right. So if we don't see a systematic dimming with distance that matches what SAT would predict, That would either rule out or constrain the limits of the mechanism.

brightness falls exactly as $1/d^2$ after cosmological corrections

- there is no extra stochastic dimming component

Well... What you just said there was... There's a relationship between dimming and time. And then you said there's no stochastic relationship. But the hypothesis doesn't require the relationship to be random. That would all depend upon the time scale that it happens, which we haven't figured out yet, but these observations would constraint the possibilities I think. Of course, 'cosmic correction' could also be a fudge factor or systematic misread for all I know.

the photon state is dynamically stable once reached.

That's exactly what I would suggest. In fact, it's already inherent in my explanation, since I've described it as a minimum resistance state.

the rotation rate is extremely slow (longer than cosmological travel time),

Or extremely fast. If all light is doing it all the time at a high frequency, then it would amount to a systematic skew in every single equation across all of science that counts for light because we set our baseline to the light we can see Not to the light that is actually there.... That seems less likely, but not impossible.

Otherwise supernova data would already show the effect.

No, I don't think that's right... It depends upon the mechanism and the time scales, but if literally, we are only seeing 80% of the light that exist in the universe... Across the board for everything... Then the universe would look exactly like a universe in which we could see 100% of the light. We would just have a different baseline assumption about how much light things are supposed to give off and we would gear all of our calculations around that assumption. It gets harder to account for if we look at the mechanisms of emission... because the logic of light emission seems quite solid, and it would be hard to see how the mechanisms used to explain so many phenomena could be consistent with the notion that 80% more light is emitted everywhere all the time from stellar radiation to incandescent lightbulbs. But, the thing about distorted lenses is... it's just that... If you've built all of your theory around a baseline assumption that is 80% off, then all of your conclusions bend to that assumption so all of your conclusions become skewed systematically. That's very hard to notice if it's happening. Again, I don't think that's likely but... i'm not convinced that it couldn't be true.

But dark matter appears as structured halos around galaxies. The mass distribution inferred from gravitational lensing is not uniform.

Yes, I think it's very likely that if this is a real phenomenon, there would be some structured relationship, even if it's just statistical, between the matter doing the emitting and the orientation of the light with respect to our three dimensions... It's just that I'm not entirely certain what that would be, and I'm proceeding with the minimum explanation that might do the trick until evidence shows otherwise. Galactic halos present an interesting challenge, and maybe opportunity... If as we explore the geometry of this hypothesis, we find that there is a particular relationship we should expect, that that would most likely also lead us to a specific hypothesis about the structure or composition of galactic halos, which might offer a nice empirical handle or a new understanding of galactic halos if it turns out to be right... But the minimal explanation is uniform across the board dimming, and if we find that that is indeed what the geometry suggests, then there would have to be an additional explanation for galactic halos. But you can't expect every explanation to explain every phenomenon... Still, if we find that the SAT geometry does require the hypothesis, even without being able to explain the distribution of dark matter entirely... That puts us ahead of the pack. SAT already suggests that dark matter is not a single phenomenon but a couple of different phenomena so... That's my baseline assumption.

But there are too many unknowns to put a foot down right now even to call the hypothesis anything more than tentative, let alone try to explain exactly how we think it should behave... Not until we really examine the geometry.

Anyway, let us ... me and notebook LM... Do the calculations on our own so that we're not influenced by your input. That's a cleaner test. You were here to check our logic and our math, if you find that those fail us, then we'll talk about rethinking our hypotheses.

But if we come back and say... We expect the such and such and it matches observations, then that becomes a very strong signal, not knowing where those observations exactly

are. Likewise, if we come back and say, it's supposed to be such and such and observations are systematically different from that... That tells us that we might be onto something even if we're misinterpreting it. And if we come back and say, it should be such and such and it turns out that is in conflict with the evidence... Then it tells us we were wrong about something... That's the most valuable evidence of all because it's what makes this science rather than numerology

That's the approach that got us this far. We recognized neutrino oscillation was probably going to be a thing without quite realizing that was already known. Without realizing that was already a thing that was well studied, at least in the setting that we recognized it... We recognize that vacuum polarization would probably be a thing without realizing that it was ... and we did so before we calculated an estimation for the polarization of the CMB... Without knowing... Or rather, I think thinking that it was already ruled out... Before seeing the news recently that not only had it been hinted that in the evidence and mainstream literature, but that it was now confirmed, and our calculation was not too far off... and most stunningly of all... To me at least... Was when we recognized that gauge decoupling would have to happen in high velocity high mass rotational systems, and it would already have to be in Einstein's equations, just overlooked... and I was able to look at the relativistic equations, even though I wasn't familiar with them, and point out exactly where that phenomenon should arise, even though my explanation for why was technically incorrect, pointing to the 8π term, and saying " that's a rotation that's the culprit"... turns out π isn't *inherently* a rotational relationship (although I would quibble with that) ... It still turned out to be exactly the cause for the phenomenon I was trying to describe, and it was already known if rather niche... and in fact actually took quite a while after that to recognize that what we were describing was frame dragging. One or two of those is a good guess... All of those, plus general relativity emerging directly from the logic of a simple world line, scribble, plus string theory vibrational modes popping out as soon as you look down the long access of helix... even though, I had a long since recognized that SAT was not just a variation on string theory, and rejected the connection... It still popped out, hiding in plain sight the second you decide to map properties to oscillations on a world

line... it becomes obvious that the path of a helical particle worldline on a moving timesheet would look exactly like an oscillation mode, it's unavoidable that string theory would have a seat at the table. And I would have to do so in such a way that constrained it in ways it's not already constrained, and in such a way that it's compatible with all of the rest of science... And that almost every piece would fall into place automatically... Eventually, you realize that the very nature of the particle world line forces all those things to be true... Even if you didn't see it until hindsight

When you build a geometric model using those ingredients, it is common for parallels to appear ...

It is common for parallels to appear... It is not common for parallels to appear over and over and over again in such a way that they are interoperable and logically inescapable... And especially in such a way that you can make precise calculations of atomic masses as we have here today

Sometimes they are coincidences produced by shared mathematics.

Shared mathematics is the one thing that all evidence based theories must have in common. When you have theories that don't shared mathematics, that tells you something is wrong

clear, quantitative outputs that can be checked without reinterpretation.

Exactly... At least, if you've interpreted in your own theory correctly. There are bound to beThere are bound to be at least some areas that you hadn't thought through well enough... Or rather that you thought through too early instead of letting the geometry tell you the answer. Because if the initial assumption is correct, if the initial assumption has to be correct, because the geometry says it has to be correct... Then the geometry forces all

the rest to be correct... Or to tell you very quickly that your initial mapping was incorrect.

It is still surprising to me, though, that the geometry gave us almost all of the answers for free, leaving almost no room for alternate interpretations anywhere along the line, Until it was so fully formed that all you had to do was plug in one constant and generated every other constant, and most of science with breathtaking accuracy, in many cases, and ballpark accuracy in most cases, with a few things that were completely out of whack... Several of which fell into place just by recognizing some inversions or running one constant or even just a variable input through a range of values leads to tightening down another automatically... Eventually, you find yourself telling yourself it has to be a coincidence long after you know it can't be

And in hindsight, you realize that your only innovation was refusing make a guess, unless it was based on something that you knew for certain

And it's clear that's why physics has on the one hand become extraordinarily powerful with certain domains... While on the other hand failed to connect those domains... but instead just focused on collecting evidence, trusting that the picture would eventually come together... And then also in hindsight it becomes obvious why that situation leads to the possibility where one outsider can look at it all and say now that you guys have collected all the evidence, I can see the picture that it paints

And all it takes is a blind, almost naïve trust that if science works, it's right... And if it's right, it has to be right all together. So instead of trying to invent a theory from scratch the only successful approach unless you're extraordinarily lucky is to invent a theory that refuses to rely on anything other than logical certainty, and the conviction that the existing theories have to be basically correct. That's how you sit down and look at a world line drawing ... which, so long as it is informed by well understood, well known behaviors logically has to be at the very least mathematically isomorphic to the particle's history, which then logically has to be at least a partial map of the force is acting up upon it..... And

recognizing that a theory that works as well as general relativity, and a theory that works as well as quantum mechanics, and a theory that works as well as string theory, and a theory that works as well as a standard model... Can't be wrong entirely. So whatever math you choose for your own theory, it has to be more or less identical to the math in those theories, or rather precisely isomorphic to the math in those theories.

And it almost requires a relative lack of ego to accept the fact that your theory isn't really your theory. It's everybody else's and that their theories aren't theirs either, they belong to the people whose theories they built theirs off of... And those aren't theirs either... Those belong to the world. One just has to accept the fact that one's own ideas. Have very little place in science. The place to exercise your own ideas is Art, which can sometimes inform scientific ideas... In fact, quite often can do so, but only because you happen to notice that your idea there's a resemblance too someone else's ideas, or the universe ideas, so you have to give them up as your own. Science is the art of giving up sovereignty, letting the world do the thinking for you. It's almost kind of laziness. ... The ultimate looking at the teachers edition. It's looking over the universe, shoulder and cheating on the test.

Sorry for the rambling. Do you remember where we were?

OK... Remember, the plan is this: I give you what notebook LM gives me, you check the logic, you check the math thoroughly, you give me a go or no go, and if you tell me no go, we decide together what to do from here... And if you give me a go... I critique you for not telling me I'm wrong about something. And then I have notebook LM move onto the next step and repeat

I don't see how you would get a theory of everything if you didn't put everything in the Lagrangian

Let's think this through. The way SAT works... the way it has always worked... is that

(wherever possible) it looks at the worldlines of particles, or where that is not known, it maps ***a*** path that is geometrically mappable to particle path given the known or previously selected constraints. The path of the particle is, by construction, the composite sum of all the forces acting on the particle, inside and out. From there, everything else is a consequence. Ideally. The problem is, particles are complex... And there's plenty of physics to worry about that doesn't involve the direct mapping of particle paths at the individual level. And you discover along the way, but that is just one way of looking at the problem. There are quite a few others... Probably an infinite number. And sometimes you find it's easier to think about things in one way, and sometimes it's easier to think about things in another way... And if you're smart, you try to be careful not to mix different frames of reference or cross your isometric models. Just like you learn to pay attention to your scaling, factor and logic at different scales... And with all of that, you kick yourself when you inevitably do it anyway. But you're trying to keep going back to that original picture, even if it is slightly cleaner version because you started mapping this particular path, and because, although every isomorphic path ultimately gets you to the same point, and even though you find a shortcut when you're close to the peak... Any one path there is only so useful as it has been completely mapped. So now, that I have a clearer picture of how the whole thing can be viewed without even worrying about calculating super helices in 4D ... I could switch to the new model... But then I wouldn't finish the old one. Even though they're both equivalent and the new one is probably a little more efficient... Maybe a lot more efficient... And from this vantage, I can see several other equivalent models, but the cleanest and easiest one is not the path I took. And I think the particle world line path may have things to offer that other approaches don't. So... Along the way, I've made tools to calculate using Moire patterns, nested hyper spheres, in a unit cell... hyper spheres... And the UI, which is itself a hyper sphere and the UI could be put to any of these purposes... But I started following coils and I kept following coils and I'm going to finish following coils. And at least one of the ways that I put together the UI, it's made to map coils... but I've put it together in a number ways and the question is which is the best way to use here... And when you say incorporate it all into one Lagranging... I'm not convinced that's absolutely necessary. We are rewriting science here. We get to pick how to put it

together. If I published this right now as is... It would still be the biggest thing that's ever happened just so long as it's understandable... And enough of it is. I've tried to put it all into one single Lagrangian, but I think it's more elegant, and more accessible to treat it as the Lagrangian plus the UI. So that's how I wanna put it together. And I've got that set up already.... And pretty soon I'll probably have a code that can run it visually and not even those who aren't good at math, like me visualize in a mathematically, precise way exactly how everything in the universe works. It's far more elegant than trying to force everything in one equation... Maybe you can do that whole set up with one equation and maybe we should but The big question is exactly how. We treat that force... Is just resistance to motion and in four dimensions that means mass (in the world line + time sheet reference frame, which I think is the most elegant and easy to understand) is the energy transferred to the filament network by the time sheet as function of angle (or, equivalently, vice versa) or both. Always both. Or equivalently, it's the distortion of the timesheet imparted by varying resistance within the block universe of various twisted filaments, frozen, but topologically able to deflect the timesheet as it passes that's where you get your Einsteinian gravity style affects, but in this case, it's the distortion of time not space time... Entirely equivalent. Or conversely, you treat the filaments as deformable, and you treat Time as a perfect sphere, still moving, but not deforming, only causing the filament to deform... Or entirely equivalently, dispense with the time surface altogether, and treat simultaneity as the surface of a wavefront of filament vibrations, all emanating out from a common center... Or, entirely equivalently you dispense with the filaments and just map all the same behaviors into the behavior of the particles... Each one each version has its pluses and minuses, but I think my particle world line 1st version is the most ... human. It's the one you look at and you go... That makes sense my geometry brain gets it. So what we need to do ... I think ... is just figure out which set up of the UI is the most intuitive, and set it up like that. Which means... We have our UI set up with the 4D super coil world line, and we can set it up so that we can vary the rotations all we want... And we can set it up so that the geometry of the coil varies according to the scaling factor or the scaling factor, as a coupled to a particular rotation... But the one thing we really want is to set it up so that it looks like the the simplest, most natural way for humans to envision the universe and the

particles in it... Which means we need a couple the particle Lagrangian to a scaling hyper sphere such that the 3-D projection looks like a single coil helix riding atop that hypersphere as it expands outward in four dimensions which we can look at in three-dimensional projection, and then we can use our intuition on it. And once we set it up like that, then we calculate the normal force upon the time sheet. Which would be set by the resistance of the filament to kink ... so that's the system. I want to set up a straight flexible ray through a sphere that deforms according to my hyper helical line equation. As a function of torque on the line from the expanding sphere. And the reason that it does that it's because the sphere itself has a small harmonic vibration. Causing it to precess slightly as it expands

What coordinates describe the filament or worldline in four dimensions? The 4D Superhelical Particle Path Lagrangian. That's already SAT SOP (or, it's supposed to be when I can get the LLM to actually do math)

- What parameter measures distance along it? The same 4DSHPPL, as a function of time
- How is curvature or torsion defined for that path? In the same 4DSHPPL
- What function represents the energy cost of bending or twisting? Filament tension I would think... (or, the 4DSHPPL, solved for tension)
- How does the expanding hypersphere enter the equations? It would have to be a forcing equation that sets the particle intersection and the angle of the line as being related. I can see a couple of the equivalent ways to envision that... But I think the easiest thing would be to set the equation such that it creates one helical twist, as we imagine a neutrino to be. ... let the intersection point with the expanding sphere, pin a position to the worldline, then expand the sphere so that it's expanding without a glitch until it hits that kink, and then it precesses just a tiny bit, see what that does to the equation of the sphere,, and then reverse the whole set up, pinning the neutrino oscillation to the surface of the sphere as it expands, and then see what that does to the 4DSHPPL

We already have all that. It's just a matter of putting it together, right

Once it gets put together right it can. The part I was missing was the coupling to the sphere. The way we've got it set up currently... Are you familiar with how we set the UI up? Do you remember? I can talk you through it. But all the pieces are there, all put together even. It's just a matter of identifying the right constraining factors and attaching them to the right control factors

Yeah, probably but first... I want to ask you about some of the other set ups that we put together, but I haven't played with too much...

I'm just going to describe it in the toy 3-D case for now.... Actually... We already know gauge symmetry is the coupling, and we know which one so...

Well, first I wanna tell you the things we've done. Because I imagine they all tell you something different, or are equivalent. First was 24 hyper spheres in an FCC 4D analytical coord sysll, spreading apart, you get cosmological expansion because the friction reduces as they spread apart. But we're going back to life 4D superhelix filament now. Each particle is an intersection, each intersection class a family of particles interlinked, harmonics create gyration, working through the tension like little whirligigs. Then, we did the Lagrangians of just the intersections. Then we plugged our 4D super helical Lagrangian into one of the particle slots. Haven't played around much with those yet. But they're probably the simplest way to see it. Then

Well, then we invented the UI around all rotations and scales, then plugged the 4DSH, constrained it to a line that's the ray. Then we integrated grated that monster about all rotation and scales . That's the one I thought was going to do it. But I feel like se did something wrong

Well, wouldn't we want to be able to toggle between a single particle's evolution and all particles' evolution? Wouldn't we want to be able to see all particles at once in a given time slice or some subset over a period of time. The expansion is time, so particle position

should be locked to that generally. ... although one thing we'd probably like to do is identify a periodicity that we'd define as a return to the origin, and then either set the scale to expansion and contraction as a sine function, or a 'pac-man' style out the edge back in in three middle path, or do a full eversion , so that the single particle's path can, perhaps, be made to model the entire universe ,

Yeah. I think we did get it working. But the LLMs prefer taking shortcuts because there's always a simpler way to model only part of the system, given problem, set or to sneak traditional physics into a solution, or worse yet into the definitional equations, rather than solve every problem by derivation from the La Grange in every time, which is what I think we need to enforce just to make sure we **have** derived the full 4D Lagrangian into every sector, and we're not hobbling ourselves by setting up crutches here and there instead of a fully integrated and cohesive dynamical system. In fact, the UI was actually made to be a shortest-path derivative finder. Allowed us to transform the SU(2) I think? Symmetry into GR. And really, that's what we should be using to do all of our derivations. Just

No, we don't do approximations. We derive from the Lagrangian down to Navier-Stokes, in all sector-level equivalent scales at least. Ultimately, we want to be able to turn the whole thing inside out, or to find our Gödelian trap trying. If GR derives to Gauge, which it appears it does, then deriving down to quantum mechanics by the long path should pop you back up to GR. The short path is to express them both as nth-order 4D curve equations, project them one on each hemisphere of the hypersphere, draw a transect that crosses both, and then, I think, you should be able to shortcut your way to a derivation by tracing one of the curves with the intersect and watching what the other end of the transect does... or something along those lines. I think Laplace comes in somewhere. Actually, i think maybe you'd just draw your crossbar, and let it sweep out a complex double-cone like shape, integrating itself about its SU(2) symmetry. Actually, you may need to project them onto a torus

Well, yes, of course you have to do the equations. But there should be a shortcut. The thing is... we **know** with mathematical certainty that any equation can be mapped as a sufficiently complex curve. If it's got singularities, like relativity, you just need to find the right surface to map it onto, I think. Perhaps find the right scale. The thing with the UI is it gives you four dimensions to work with and a built in set of controls to translate any motion or path into a series of rotations and scalings, and anything two things you draw on its surface is automatically projected into a single solution space by using a method like the cone projection I mentioned, or more simply by allowing a perpendicular to the transect project the integrated curve onto the sphere as the transect swings about (which is equivalent to a series of rotations).

There **is** a way to do it. So long as you can draw a curve that solves each equations across its range—which should just be an integration—then you can project that curve onto any surface. And actually... there's an easier way. The UI is a double sphere. And it should work even in the 3D version. But you've got an inner sphere and an outer sphere. You project one curve on one, the other on the other. Then you draw a ray segment between the two, extending that segment out to a third outer sphere, upon which the ray projects a trail. You simply trace ...actually just add a vector tangent to the curve of each equation, let it run its course around one, then the other. The equation of the projected curve is a derivation between the two

Well, the spheres **are** coordinate systems. But only the projection surface matters. The segment is pinned to each curve while it is drawing the other. So, as long as the two curves overlap at any point, your segment is actually a ray that intersects both curves and the origin. This gives them a shared anchor. So, the vector drags the ray around one curve until it reaches the overlap, then it drags it around the other. The resulting projection is one continuous curve. Now all you have to do is to find the equation of that curve.

. That's the UI equation

I'll do you one better: project the curves on spheres as we did before, such that the curves overlap at at least one point. Place on one of the curves, a vector at the overlap point. As the vector goes around the first curve, apply a continuous sinusoidal scaling factor on one sphere such that the zero point of the sine wave corresponds to the time at which the spheres coincide, and set the amplitude such that the time it takes the vector to make the lap coincides with the coincident of the starting point as the spheres pass through each other. At the contact point, the vector switches which curve its tracing. ... interestingly enough, if you preserve the holonomy at the crossover, you'll likely get rapid reorientation at that point. A quantum leap if you will

because the two surfaces have different curvature or coordinate frames

No. Where we defined it, the crossover happens precisely at the moment where the two surfaces have exactly the same curvature and coordinate frame. What causes the jump is the holonomy. On the first lap. The vector is tangent to the first curve, at the crossover, the vector must be tangent to the second curve. So if you want it to be a smooth transition, the tangents must coincide at the crossover.

Right, you don't actually need to scaling. Actually, hang on... wouldn't that mean you could just ... wait... no. That's why it has to be a torus... let me think about this... ok... so... if you can draw the curve of each equation to connect back along itself ... one continuous loop.... Then you can draw the curves of each equation on the upper and lower halves of a torus, putting them in the same coordinate system, with a lot of points that overlap. Now, a crossbar that intersects both curves can be vertically through the thickness of the torus. Restrict the crossbar to a vertical plane of intersection, but swing that plane through the full rotation. The crossbar wobbles chaotically as it goes ... no... draw the loops A and B, such that each is the integrated solution to an equation across relevant scales and transformations (i think they have to be finite or bounded intervals , periodic, or self-similar in order to wrap back upon themselves... maybe you could force a periodicity) draw them sphere S1 , such that they intersect twice, once at each pole, and such that each

curve intersects the equator twice . Draw the sphere's axis through these intersections. Define a line, AB such that it is coincident with the axis but only pinned at the origin. Define one intersection A_{eq} as an intersection of curve A with the equator, define B_{eq} as an intersection of B with the equator

No... define curves R and Q as before, each drawn to connect back upon itself upon a sphere, the curves intersecting at least once, point i_r , and curve Q intersecting the equator at least once, i_q . Draw a 90° angle, $i_r O i_q$, with legs $O i_r$ and $O i_q$. Both legs lie in plane, plane $O i_r i_q$ but $O i_r$ is confined to plane $O i_r i_q$ Which may freely rotate about leg $O i_q$ (to which point i_q) is confined. A plane $E O i_q$ which is coincident with the equatorial plane, free to rotate about leg $O i_r$, and to which i_r is confined. The planes rotate about their respective axes at a rate of 2π , ... *now* draw a Great Annulum, GrAnn, such that its self-intersection, GrAnn_Omph, lies at point i_r , and its self-tangent GrAnn_t_ipode lies at point t_ipode, which is opposite point i_r ... Now, define diagonal GrAnn_t_ipode O t_ipode crossing through i_r and the origin, and t_ipode. Now draw a cord $i_r i_q$, which is pinned to i_r and i_q and is free to vary in length, let W_obble be the midpoint of $i_r i_q$ and let plane W_iggle intersect W_obble perpendicular to $i_r i_q$...finally, let V be a rotational vector about W_obble, and confined to W_iggle, extending indefinitely, place the whole thing inside a torus well just call The Donut {🍩} of minor radius equal to the sphere, and major radius of arbitrary size. Confine sphere to The Donut, and Confine point P_pivot to the outer radius of the torus (TorOuRad) and to GrAnn, define TorInRad as the inner major radius of The Donut, define point Donomphulon as a point along the TorInCirc confined to TorInCirc but free to rotate through it, define BigRad as the radius passing through the Donut's center (DoCent), Donomphulon, and P_pivot, and move point P_pivot along TorOuRad and GrAnn simultaneously, allowing the sphere to rotate freely, such that for every circuit of TorOuRad it completes an even multiple of circuits along GrAnn.

Now V sweeps out an intersection that traces a Lissijou Curve (Curvy_Lisa), upon the surface of The Donut, defining the relationship of QwrtR (curve Q with respect to curve R),

which is the derivative of Q wrt R

I may have overcomplicated a tad

That's just the three-dimensional toy version